

Running Prosthetic Blade

FEA-led structural optimisation

A lightweight blade concept developed and validated through static, modal and fatigue analysis.



CICI SONG

Product Design Engineer | Fusion 360 + ANSYS 2025 R2

Design intent

Create a curved spring-like running blade that stores and releases elastic energy while remaining lightweight, durable and dynamically stable.

< 1 kg

CFRP MASS TARGET

> 1M

FATIGUE CYCLES

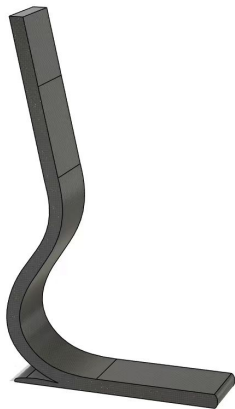
> 50 Hz

FUNDAMENTAL FREQUENCY

< 1%

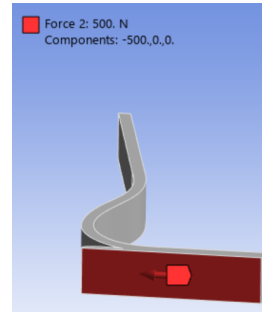
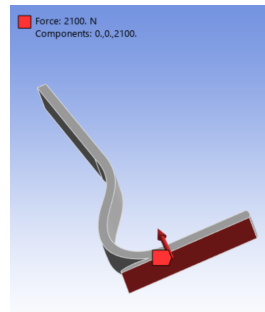
MESH CONVERGENCE

CAD + MATERIALS



Initial geometry modelled in Fusion 360.

BOUNDARY CONDITIONS



Fixed socket interface

2,100 N normal load

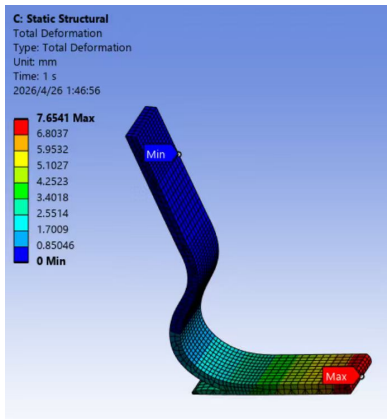
500 N backward friction

Evaluated in CFRP and Aluminium 2014-T4.

VERIFICATION METHOD

- Quadratic elements with local refinement at high-gradient regions.
- Coarse, medium and fine meshes compared.
- Static structural, modal and fatigue analyses.
- Final mesh accepted when successive changes fell below 1%.

INITIAL CFRP - FINE MESH



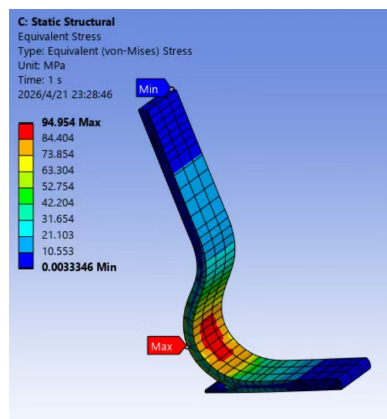
DEFORMATION

7.65 mm

MAX DEFORMATION

69.37 Hz

FIRST MODE



STRESS

101.76 MPa

PEAK STRESS

0.81 kg

EST. MASS

WHAT THE BASELINE REVEALED

Critical region

Peak stress consistently formed at the inner curved transition, identifying the geometry that required refinement.

Material trade-off

- CFRP: lower peak stress, higher first natural frequency and roughly half the mass.
- Aluminium: slightly lower deformation, but lower dynamic performance and higher mass.

Baseline comparison

54.54 Hz

ALUMINIUM FIRST MODE

1.58 kg

ALUMINIUM MASS

Decision

Retain CFRP as the preferred material and improve the curved transition to reduce local stress concentration.

Both materials met the initial fatigue and >50 Hz criteria.

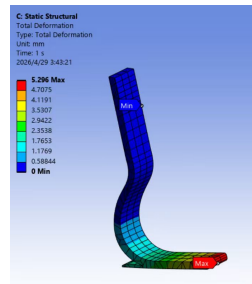
ITERATED GEOMETRY



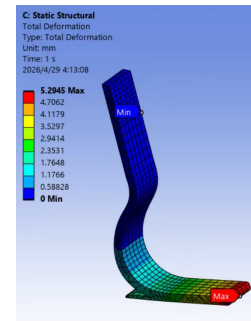
Three targeted changes

- Increased inner arc radius.
- Thickened the high-stress bending region to 18 mm.
- Shortened the front foot while maintaining spring behaviour.

VALIDATED CFRP RESULTS



5.30

mm
deformation

76.26

MPa stress

71.12

Hz first
mode

0.80

kg mass

MEASURABLE IMPROVEMENT

-25.1%

PEAK STRESS

-30.7%

MAX DEFORMATION

+2.5%

FIRST NATURAL FREQUENCY

-1.2%

ESTIMATED CFRP MASS

The iteration reduced stress and deformation without sacrificing the lightweight target or fatigue margin.

Final material recommendation

CFRP

Best overall balance of mass, stress response and vibration performance.

The iterated design remained below 1 kg, exceeded the 50 Hz target and satisfied the simplified fatigue criterion with a safety factor of 3.

FINAL CFRP PERFORMANCE

0.80 kg

ESTIMATED MASS

76.26 MPa

PEAK STRESS

5.30 mm

MAX DEFORMATION

71.12 Hz

FIRST MODE

228.78 MPa < 717 MPa

Safety-factor check under stated assumptions

WHAT THIS PROJECT DEMONSTRATES

- Parametric CAD modelling and design iteration in Fusion 360.
- Mesh-convergence checking and defensible FEA setup.
- Static stress/deformation, modal and fatigue interpretation.
- Material selection based on quantified trade-offs.
- Turning simulation evidence into clear geometry changes.

TOOLS

Fusion 360 | ANSYS 2025 R2 | Engineering analysis